EUROPEAN FORMAT FOR CURRICULUM VITAE

ALTOMARE Francesco

Big Data Engineer

PERSONAL INFORMATION

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Nationality	Italian
Date of Birth	17/04/1999

WORK EXPERIENCE

01/2026 – present
Contrada Cutura, Via Spagna, 240, 87036 Rende CS NTT DATA Italia
Sector: Artificial Intelligence (cross-sector)

AI Engineer – AI Toolkit

Since January 2026, in parallel with my ongoing project in the insurance sector, I have been involved in the design and development of an AI Toolkit plugin built to be used by everyone. The toolkit includes a Skills Marketplace that collects agents, skills and plugins ready to be installed on the most widely used AI harnesses, such as Claude Code and Codex. Its most innovative component is a library of AI-based workflows designed for different goals, which guide the user step by step and allow even people with no prior AI experience to benefit from these tools. In this project I work hands-on with the full stack of modern agentic AI technologies: designing custom agents and multi-agent workflows, building skills and plugins, configuring and integrating MCP (Model Context Protocol) servers, using hooks to automate and control agent behaviour, and adopting graph-based memory to give agents persistent, structured context.

Key technologies: Claude Code, Codex, AI Agents, Multi-Agent Workflows, Skills, Plugins, MCP (Model Context Protocol), Hooks, Graph Memory, Python

17/12/2026 – present
Contrada Cutura, Via Spagna, 240, 87036 Rende CS NTT DATA Italia
Sector: Insurance

Big Data Engineer

I am currently working on a project in the insurance sector as Senior Data Engineer, with the responsibility of designing, developing, and maintaining high-performance ETL pipelines. The solutions are mainly implemented in PySpark and Python (with the Polars library), or by leveraging native Azure Synapse services. In this context, I have introduced an Agentic Development approach at the enterprise level, designing skills and custom agents developed ad hoc with the help of Claude Code and Codex to support technical design, assisted implementation, and solution validation. The infrastructure is hosted on Microsoft Azure, where I work with Azure Data Factory (ADF), SQL Server, and Synapse Analytics, as well as supporting services for orchestration, monitoring, and data management. Data is stored on Azure Data Lake Storage Gen2 (ADLS Gen2) following a Delta Lake approach: I design and manage Delta tables, implementing incremental loads through append and merge/upsert patterns (DeltaTable.merge) for deduplication, schema evolution with mergeSchema, and partition-aware writes to optimize performance and storage. On the agentic side, I have built multi-agent workflows and MCP configurations that support colleagues across different domains, such as data engineering and cybersecurity, bringing AI-driven automation into the team's daily work. This experience is allowing me to consolidate my skills in designing cloud data-driven architectures and in optimizing ETL flows in highly scalable environments.

Key technologies: PySpark, Python (Polars), SQL, Delta Lake, Microsoft Azure (Synapse Analytics, Azure Data Factory, ADLS Gen2, SQL Server), Claude Code, Codex, AI Agents, MCP

01/2025 – 06/2025
Cosenza, Italy
Sector: Telecommunications

Big Data Engineer

I was involved in a migration project for a major telecommunications company, aimed at transferring the entire data infrastructure and ETL processes from an Oracle/DWH on-premise system to Google Cloud Platform (GCP). I worked in the team's solutioning area, contributing both to the development and migration of data flows, and to the definition of technical solutions to overcome constraints arising from the infrastructure paradigm shift. Activities included adapting complex logic (PL/SQL) to GCP-native environments, rewriting ETL jobs in Python and SQL, developing DAGs in Airflow for optimized orchestration of all migrated jobs, and using technologies such as Apache Spark for distributed processing.

Key technologies: Python, SQL, PL/SQL, Apache Spark, Google Cloud Platform (BigQuery, Cloud Composer, etc.)

06/2024 – 12/2024
Cosenza, Italy
Sector: Telecommunications

Big Data Engineer

As team leader, I manage, together with my team, the design and development of a Google Cloud Platform (GCP) infrastructure to replicate a series of ETL jobs developed in SAS and Oracle. We use services such as BigQuery, Composer (Airflow), Cloud Storage and Cloud Build, while the adopted programming languages are SQL and Python.

04/2024 – 06/2024
Cosenza, Italy
Sector: Telecommunications

Big Data Engineer

During this period I was engaged in designing and developing an infrastructure on Cloud GCP, for the application of SQL logic that would allow the calculation of surveys, intended for the various clients of the provider I worked for. The services I used and configured are: BQ, Composer (Airflow), Cloud Storage, Cloud Build. Programming languages used: SQL and Python.

02/2024 – 04/2024
Cosenza, Italy
Sector: Entertainment

Big Data Engineer

During this period, I worked on developing a pilot project designed to migrate several ETL processes from an on-premise (SaS) environment to Google. Mainly, with this project I used Composer and Airflow for the orchestration of the various processes that together made up the various SaS ETL jobs, together with BigQuery for the migration of SQL scripts previously developed to run on SaS. The fundamental aspect is that to fully migrate these ETL jobs, I developed custom operators (Python operator) to use on Airflow, which had numerous tasks, as well as an innovative framework that allowed creating or modifying DAGs with extreme ease by compiling a JSON file.

09/2023 – 02/2024
Cosenza, Italy
Sector: Multi-sector

Big Data Engineer

During this period, I worked on developing an innovative project with the aim of acquiring data from multiple sources, such as: APIs, on-premise DBs, Firebase and Google-Ads, and using them to create KPIs that would allow tracing the history of a user's activities on the various applications of a potential customer. During this period I was able to work with various Google technologies, such as: Composer and Airflow for orchestrating Dataflow jobs (batch and streaming), and for execution through custom operators (Python operator), of various operations such as: launching parameterized SQL scripts on BQ, sending messages on pub/sub to trigger other procedures, as well as Dataflow jobs. Furthermore, I was able to work with BigQuery for developing SQL queries, Cloud Build and Cloud Run for developing Python functions to acquire data from APIs, or provide APIs for data push, Stack Driver for log acquisition, to use for triggering other Cloud Run functions.

07/2022 – 09/2023
Cosenza, Italy
Sector: Entertainment

Big Data Engineer

I worked for about a year in the processing team for an important client in the entertainment sector. My role was to manage data coming from the ingestion area through the services provided by the Google Cloud platform. Specifically, I focused on structuring data and extracting KPIs through SQL/PLSQL procedures used by Business Intelligence. In doing so, I used, in addition to SQL, also Python and Java.

Google services used: Composer (Airflow), Dataflow, Cloud Logging, Pub/Sub. Also the standard Microsoft services, Grafana and Collibra.

03/2022 – 07/2022
Cosenza, Italy
Sector: IT

Big Data Engineer

During this period, I worked with my team on creating a connector-based platform, tasked with migrating company data from an on-premise context to the cloud GCP, through the use of the Apache Beam framework and the Java programming language. Everything was divided into 3 levels: 'LVL 1' (ROW) where unstructured data saved in CSV format inside a GCS bucket are transformed into semi-structured data and saved in another bucket in '.avro' format. 'LVL 2' where data is written on TS tables in BQ. 'LVL 3' where data moves from TS to SS tables and is enriched for KPI calculation. The first connector 'csv-connector' was designed to read .csv files from GCS buckets, transform them into semi-structured '.avro' files and deposit them in another GCS bucket. From there, the second 'dvse-connector' would read these '.avro' files, and the semi-structured data contained therein were then written to Big Query tables. Then through the use of doc scripts, such data were used to calculate KPIs useful for the business. During the development process, the most sophisticated methods of data tracking (Tracking table), data quality control (DLQ table) and masking of sensitive data were adopted, through the use of a library created by us that was based on KMS and Secret Manager services from Google. The whole was implemented using the languages: Java8 along with the ApacheBeam framework, Python and SQL. Two execution modes were planned: 'Initial Load' where data is loaded in bulk, and 'On-Going' where data is instead processed in delta, recovering even failed runs from previous days. The entire process was inserted within Composer, which orchestrates the various jobs for scheduled and sequential execution.

03/2023 –
Cosenza, Italy
Sector: IT

Big Data Engineer

I am working on developing (through the use of programming languages: Java8 along with the ApacheBeam framework, Python and SQL) a connector that will allow data migration from on-premise source databases to the GCP platform. Specifically, such data can be saved in GCS buckets in different formats (.csv, .avro...), or can be inserted directly into Big Query structures. Like all the other connectors described above, this one is also equipped with resilience and data tracking methods.

21/03/2022 – 17/12/2025
Elmiro Tavolaro, VIA MARCO POLO 3 - 87036 Rende (CS)
Sector: Software Development

Big Data Engineer

VT-Solutions and Consulting

EDUCATION AND TRAINING

2013 – 2017	IIS IPSIA Marconi CS – Technical and Scientific High School, ITE Guarasci Rogliano
2017 – 2022	University of Calabria (UNICAL) – Computer Engineering **Degree in Computer Engineering**
02/2024	Google Cloud Professional Data Engineer Certification
06/2026	Microsoft Azure Fundamentals (AZ-900) Certification
07/2026	Databricks Certified Data Engineer Professional (valid until 2029)
08/2026	Google Cloud Professional Data Engineer Certification – Renewal (valid until 2027)
In progress	Claude Professional Architect Certification – in preparation
Main subjects	Knowledge of the Cloud world and in particular GCP, programming in various languages, problem solving.

LANGUAGES

Mother tongue	**Italian**
Other language	English
Reading	High
Writing	High
Oral expression	High

PERSONAL AND RELATIONAL SKILLS

I have developed good relational skills, thanks to my years of university study and to my current job, which requires collaboration with members of my own development team. The personal skills I have acquired are due to my desire to engage and my curiosity.

ORGANISATIONAL SKILLS

I received an excellent organisational mindset from my university course, and not only, because it is in my nature to be very disciplined. This allows me now to organise my life to be as productive as possible. I usually use a to-do list for the activities to be carried out, working progressively.

TECHNICAL SKILLS

I have developed numerous personal projects in Java, JavaScript, Angular, Spring, etc., and I have also acquired a discrete competence in the Networking (IP) field, and management of Linux-based OS and obviously numerous skills in the Big Data field. The technologies I am most familiar with are:

- Java, JavaScript, TypeScript, Python, CSS, Angular, Spring, SpringBoot, Spark
- Bash commands for managing Linux-based systems, Apache Beam, Air-Flow
- GCP services (BQ, Composer, Dataflow, Cloud Scheduler, Cloud Logging, Container Registry, Artifact Registry, Pub/Sub, IAM, Cloud Run, Cloud Function, BigTable)
- Azure services (Azure Data Factory, Azure Synapse Analytics, Azure Synapse native services, SQL Server on Azure)
- SQL, GIT, Spark, Databricks
- Claude Code, Codex, Skills, Plugins, AI Agents & Multi-Agent Workflows, MCP (Model Context Protocol), Hooks, Graph Memory, Agentic Development

In summary:

- Software development for creating data pipelines and beyond.
- Configuration and management of GCP and Azure services.
- OS management (Linux, Windows, and Mac).
- Use of monitoring systems such as Grafana.

OTHER SKILLS AND COMPETENCIES

Architecture & Design

Cloud Architectures, Design Patterns, Web and Desktop Applications, Hybrid Architectures

Development & Factory

Java, Python, Scala, CSS, JavaScript, TypeScript, Bash, Apache Beam, Angular, Spring Boot

IDEs: IntelliJ IDEA, PyCharm, Visual Studio, Claude Code, Codex

Database
MySQL, PostgreSQL, Big Query

Cloud / Security & Infrastructure
GCP: Big Query, Composer, Dataflow, Container Registry, Cloud Logging, Secret Manager, Cloud Storage
Azure: Azure Data Factory (ADF), Azure Synapse Analytics, Azure Synapse native services, SQL Server on Azure
Grafana, Collibra. KMS, Secret Manager, Data Catalog.

Working Methodologies
Agile (Scrum/Kanban ...), GitHub, GitLab, Maven

Sector
Big Data

DRIVING LICENCE

Licence Category B

I authorise the processing of my personal data contained in the CV pursuant to art. 13 of Legislative Decree 30 June 2003 n. 196 – 'Code regarding the protection of personal data' and art. 13 GDPR 679/16 – 'European regulation on the protection of personal data'.